Neuro-Symbolic and Cognitive AI Chatbot for Adaptive Stress Regulation Among Undergraduates

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Introduction

- Increasing academic pressure and emotional stress among university students
- Limited availability and accessibility of timely, personalized support systems
- Growing potential of AI to deliver adaptive, real-time assistance
- Proposal of an intelligent chatbot for personalized emotional support
- Focus on enhancing student well-being, engagement, and decision-making

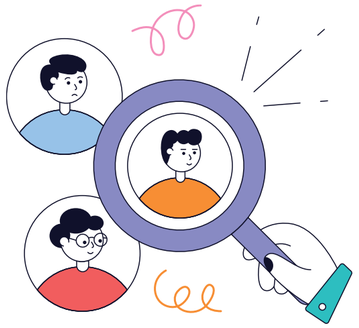
The Problem

- Rising stress and cognitive overload in university students.
- Leads to harm in academics and overall well-being.
- Existing and Traditional solutions are not scalable and have limited access.

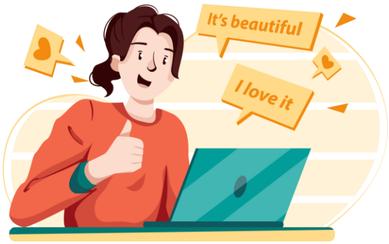
Social and Research Impact of the Solution

Social Impact

Early mental health support for students



Stigma-free & personalized

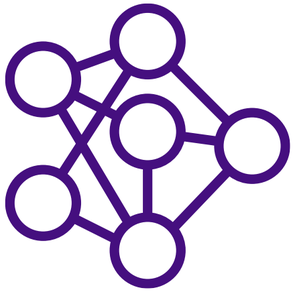


Better academics & retention

Research Impact

Explainable neuro-symbolic AI

Cognitive load detection via typing behavior and voice inputs



Stronger continual learning

Adaptive multimodal support

Design Excellence of the Overall Solution

- The proposed system demonstrates design excellence by employing a **human-centered, privacy-first AI architecture** that delivers personalized academic and emotional support through real-time, on-device, and secure data processing.

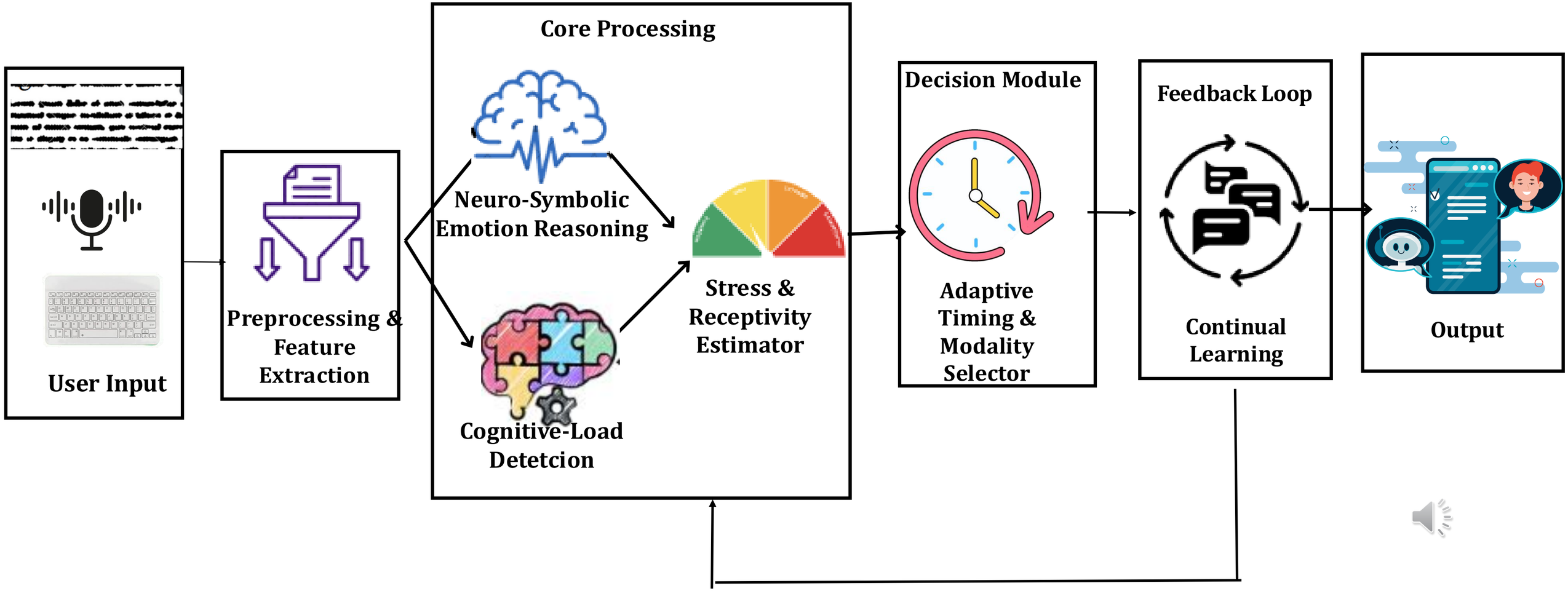
- Its **modular, scalable, and adaptive design** integrates intelligent conversational models, contextual analysis, and continuous user feedback to enable responsive, empathetic, and personalized interactions across diverse student needs and usage scenarios.

- The system further achieves design excellence through **efficient resource utilization and ethical AI principles**, ensuring high-quality assistance while maintaining transparency, data security, and user trust.

Who Will Be Benefited by the Solution

- **Undergraduate Students**
- **University Counselors & Mental Health Professionals**
- **Academic Institutions & Administrators**

Overall System Diagram



Neuro-Symbolic Emotion Reasoning Engine

Data Science

- **Sub-Problem**
  - Existing stress detection systems are often black boxes, providing little explainability, and fail to support undergraduates dealing with academic and personal stress effectively.
- **Solution**
  - A Neuro-Symbolic AI system that combines voice and text analysis with a knowledge graph of stress triggers, symptoms, and coping mechanisms to detect stress accurately and provide explainable insights for undergraduates.
- **User Requirements addressed by Solution**

  •Real-time Stress Detection: Users can speak or type, and the system captures stress instantly from voice and text.

  •Explainability: Provides insights into *why* the user is detected as stressed using a knowledge graph with evidence, stress triggers, and symptoms.

  •Context Awareness: Tailored for undergraduates, considering multi-task and academic stress patterns.

  •Multi-Modal Input: Supports both text and voice input for flexibility.

- **Design Excellence / Contribution:**
  - **Neuro-Symbolic AI Integration:** Combines neural networks for stress detection with symbolic reasoning via a knowledge graph for explainability.
  - **Multi-Modal Architecture:** Processes both voice (Vosk + audio feature extraction) and text (XGBoost / transformer models) seamlessly.
  - **Explainable Predictions:** Uses a structured knowledge graph with nodes like *Evidence*, *StressTrigger*, *Symptom*, and *RiskModifier* to provide rationale behind stress scores.
  - **Accent & Context Adaptation:** Fine-tuned models for Sri Lankan undergraduate accents and typical academic stress contexts.
- **User Feedback on Prototype:**
  - **Engaging & Interactive:** Students appreciated the ability to talk or type and see instant stress detection.
  - **Clarity & Insight:** Users found the explanation via the knowledge graph helpful to understand *why* they were flagged as stressed.

Cognitive Load Detection via Passive Behavioral Signals

Data Science

Sub-Problem:

- Lack of objective, real-time methods to detect student cognitive overload.

Solution:

- AI-based cognitive load estimation using multimodal behavioral and interaction signals.

User Requirements addressed by Solution

- Real-time cognitive load monitoring during learning and assessments.
- Non-intrusive and passive data collection.
- High accuracy with minimal false detections.
- Privacy-preserving and secure data handling.
- Clear and actionable feedback for self-regulation.

Design Excellence / Contribution:

- Multimodal feature fusion for robust estimation.
  - Combines behavioral and interaction signals for more reliable cognitive load estimation.
- Adaptive thresholds personalized to individual baselines.
  - Personalizes cognitive load boundaries based on individual user baselines.
- Lightweight models optimized for edge devices.
  - Optimized for real-time inference within device memory and computation limits.

User Feedback:

- Detected load aligned with perceived mental effort.
  - Users felt detected load levels matched their actual mental effort.
- Preferred non-intrusive monitoring over manual self-assessments.
  - Passive sensing was preferred over frequent self-report questionnaires.
- Cognitive load indicators helped improve pacing and focus.
  - Cognitive load indicators helped users manage pacing and take breaks effectively.

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Emotion – Aware Continual Learning Engine with Personalized Trends Visualization

Data Science

Sub-Problem

Most chatbots are static and no long-term emotional memory caused the issue of catastrophic forgetting.

Solution

Emotion-Aware continual learning model with emotional memory database and dashboard for trends visualization.

User Requirements Addressed by the Solution

- Real-time Emotion Detection from user text during conversations
- Adaptive Learning that updates emotions over time without forgetting
- Personalized trends using long-term emotional memory
- Emotional Transparency through stored emotion trends and visualization

Design Excellence / Contribution

- **Continual Learning with EWC** - Session-wise BERT training with Elastic Weight Consolidation to reduce catastrophic forgetting.

- **Emotional Memory Mechanism** - Persistent storage of user emotions with timestamps and confidence to support personalization.

- **Modular and Explainable Architecture** - Clear separation of learning, memory, and response components, enabling transparency and future trend visualization.

User Feedback on Prototype

- **Personalization and Continual Learning** - Users noticed the chatbot adapted over time, remembering past emotions without forgetting previous patterns.

- **Transparency and Trust** - Displaying emotion trends and memory-based interactions increased confidence in the system's understanding.

Adaptive Intervention Timing & Modality Selector

Data Science

Sub-Problem:

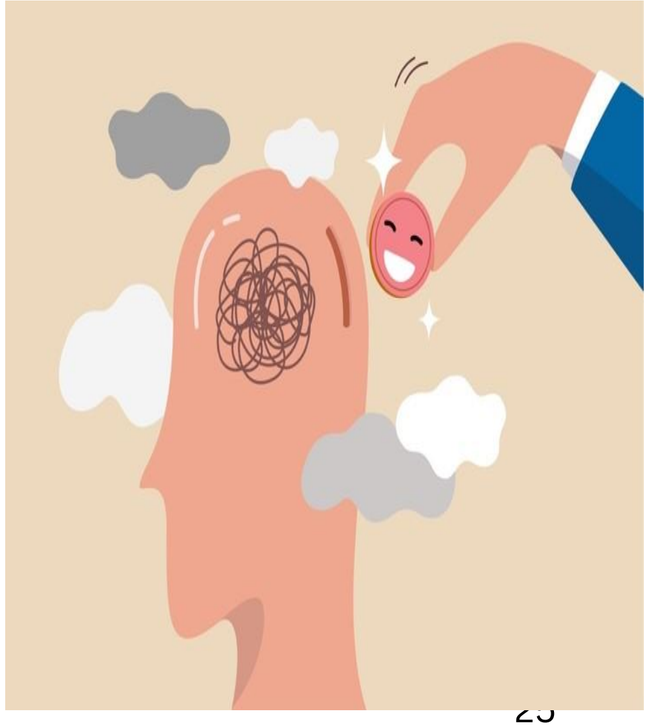
Existing stress detection systems focus mainly on identifying stress but fail to decide **when** and **how** to deliver support. As a result, interventions are often poorly timed, intrusive, or ineffective when users are not receptive.

Solution:

An **Adaptive Intervention Timing and Modality Selector (AITMS)** that operates as a decision layer within the system. The module evaluates user receptivity and sustained stress levels, then dynamically selects the most suitable intervention format (text, audio, or video).

User Requirements Addressed by the Solution

- **Non-Intrusive Support**
  - Interventions are triggered only when the user is receptive (face presence detected).
- **Timely Stress Intervention**
  - A 30-second stress aggregation window ensures stable and meaningful decisions.
- **Adaptive Support Delivery**
  - Text, audio, or video interventions are selected dynamically based on stress level.
- **Reduced Repetition**
  - Feedback-aware logic avoids repeatedly delivering the same type of intervention.



Design Excellence / Contribution

- **Decision-Centric Architecture**
  - Focuses on *when* and *how* to intervene, not only on stress detection.
- **Receptivity-Aware Logic**
  - Uses real-time face presence as a proxy for user availability.
- **Time-Windowed Stress Validation**
  - Reduces false triggers caused by brief or accidental facial expressions.
- **Modular & Integration-Ready Design**
  - Easily integrates typing speed and voice stress scores from teammate components.

User Feedback on Prototype

- **Reduced Intrusiveness**
  Users are not interrupted when not facing the camera.
- **Stability vs. Responsiveness**
  The time-window approach prevents frequent false alerts.
- **Usability**
  Text, audio, and video outputs are simple and easy to understand during the demo.

Commercialization

- **Investment / Cost**
  - **Labor:** Undergraduate research team (Python, ML, NLP)
  - **Consultation:** Psychology professor for expert consultation
  - **Materials:** Cloud GPUs (e.g., Colab Pro)
- **Recovery of Investment**
  - University licensing
- **Target Market & Customer Profile**
  - Universities & student populations
- **Value Proposition**
  - **Explainable AI:** Transparent, human-readable emotional insights
  - **Cognitive Load Detection:** Tracks typing/behavioral cues
  - **Continual Learning:** Adapts without “forgetting” (EWC)
  - **Adaptive Interventions:** Delivers timely, personalized support (text, video, audio)